

MATP32

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Lecture 1

Remark 1. In this course, we denote the Lebesgue measure of $A \subseteq \mathbb{R}^d$ as $|A|$. For integrals against the Lebesgue measure, $d\lambda(x)$ will be written as dx .

Definition 2. The space of *locally integrable functions* is defined as

$$L^1_{loc}(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \rightarrow \mathbb{C} \mid f \text{ measurable, } \int_K |f(x)| dx < \infty \forall K \subseteq \mathbb{R}^d \text{ compact} \right\}.$$

Definition 3. For $f \in L^1_{loc}(\mathbb{R}^d)$ and $x \in \mathbb{R}^d$, define

$$(M_C f)(x) := \sup_{r>0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y)| dy$$

and

$$(Mf)(x) := \sup_{\substack{B \ni x \\ B \text{ open ball}}} \frac{1}{|B|} \int_B |f(y)| dy$$

as functions from $\mathbb{R}^d$ to $[0, \infty]$. These are called the *centered maximal function* and *maximal function* respectively.

Proposition 4. For $f \in L^1_{loc}(\mathbb{R}^d)$, $x \in \mathbb{R}^d$, $M_C f(x) \leq Mf(x) \leq 2^d M_C f(x)$.

Proof. The first inequality follows immediately from the fact that the set of open balls centered at x is a subset of the set of open balls containing x . For the second inequality, note that if $x \in B$ for some open ball B with radius r , then $B \subseteq B_{2r}(x)$ as all points of B are at most $2r$ away from x . Thus,

$$\frac{1}{|B|} \int_B |f(y)| dy \leq \frac{1}{|B|} \int_{B_{2r}(x)} |f(y)| dy = \frac{2^d}{|B_{2r}(x)|} \int_{B_{2r}(x)} |f(y)| dy,$$

and the corresponding inequality must hold for the suprema. $\square$

Remark 5. Clearly, $\|Mf\|_\infty \leq \|f\|_\infty$, so M can be seen as an operator from $L^\infty(\mathbb{R}^d)$ to $L^\infty(\mathbb{R}^d)$. For L^1 , however, the analogous result is false. In fact, we have that if $f \in L^1(\mathbb{R})$ is non-trivial (i.e. $|\{x \mid |f(x)| > 0\}| > 0$), then

$Mf \notin L^1(\mathbb{R}^d)$. To show this, note that by our assumption, there must exist some $R > 0$ with $\int_{B_R(0)} |f(y)| dy > 0$. Then, for $|x| > R$, we have that

$$\begin{aligned} \frac{1}{|B_{R+|x|}(0)|} \int_{B_{R+|x|}(0)} |f(y)| dy &\geq \frac{1}{|B_{R+|x|}(0)|} \int_{B_R(0)} |f(y)| dy \\ &\geq \frac{1}{|B_{2|x|}(0)|} \int_{B_R(0)} |f(y)| dy \\ &\geq \underbrace{\frac{1}{x^d}}_{\text{too big to be integrable}} \underbrace{\frac{1}{|B_2(0)|} \int_{B_R(0)} |f(y)| dy}_{\text{non-zero constant}}, \end{aligned}$$

Lemma 6 (Vitali covering lemma). *Let $\{B_\ell\}_{\ell=1}^N$ be a finite collection of open balls in $\mathbb{R}^d$. Then, there is a finite subcollection $\{B_{\ell_j}\}_{j=1}^M$ which is pairwise disjoint and satisfies*

$$\left| \bigcup_{\ell=1}^N B_\ell \right| \leq 3^d \left| \bigcup_{j=1}^M B_{\ell_j} \right|.$$

Proof. Without loss of generality, the balls are ordered by size, i.e. $|B_1| \geq |B_2| \geq \dots \geq |B_N|$. Define the subcollection recursively by $\ell_1 = 1$ and

$$\ell_{j+1} = \min \left\{ k \in \{\ell_j, \dots, N\} \mid B_k \cap \left(\bigcup_{i=1}^j B_{\ell_i} \right) = \emptyset \right\}$$

until that set is empty. In other words, we pick the largest ball, and keep picking the largest unpicked ball which is disjoint from the ones we've picked so far. Clearly, our chosen subcollection is pairwise disjoint. Now, it is enough to show that that

$$\bigcup_{\ell=1}^N B_\ell \subseteq \bigcup_{\ell=1}^M \underbrace{3B_{\ell_j}}_{\text{triple radius}}.$$

To this end, note that by our algorithm, all balls on the left hand side are either chosen, or intersect non-trivially with a larger, chosen ball. Moreover, it is a geometric fact that if two balls intersect non-trivially, then tripling the radius of the larger will make it fully contain the smaller (this can be shown using e.g. the triangle inequality). Taking the Lebesgue measures on both sides and factoring out the 3 gives the result. $\square$

Theorem 7. *For $f \in L^1(\mathbb{R}^d)$ and $a > 0$, we have that*

$$|\{x \in \mathbb{R}^n \mid Mf(x) > a\}| \leq \frac{3^d}{a} \int |f(y)| dy.$$

Proof. Denote the set in the left-hand side of the claim by E_a . Without loss of generality, $|E_a| > 0$. By the regularity of the Lebesgue measure, $|E_a| = \sup_{\substack{K \subseteq E_a \\ K \text{ compact}}} |K|$, so it is enough to show the inequality for all compact subsets

of E_a . Fix such a K , which is without loss of generality non-empty. Then, for each $x \in K$ let B_x be an open ball containing x such that

$$\frac{1}{|B_x|} \int_{B_x} |f(y)| dy > a$$

(B_x exists because $x \in K \subseteq E_a$). Then, $K \subseteq \cup_{x \in K} B_x$, so by the compactness of K there must exist a finite subcover $x_1, \dots, x_N$ with $K \subseteq \cup_{i=1}^N B_{x_i}$. From these, use Lemma 6 to extract a subcollection $B'_1, \dots, B'_M$ with

$$|K| \leq \left| \bigcup_{i=1}^N B_{x_i} \right| \leq 3^d \left| \bigcup_{i=1}^M B'_i \right| = 3^d \sum_{i=1}^M |B'_i| \leq 3^d \sum_{i=1}^M \frac{1}{a} \int_{B'_i} |f(y)| dy \leq \frac{3^d}{a} \int |f(y)| dy.$$

□

Lecture 2

Definition 8. Let $(X, \mathcal{A}, \mu)$ be a measure space, $0 < p \leq q \leq \infty$. Then, define $L^p(X) + L^q(X)$ as $\{f + g \mid f \in L^p(X), g \in L^q(X)\}$.

Lemma 9. Let $(X, \mathcal{A}, \mu)$ be a measure space, $0 < r < p < q \leq \infty$. Then, $L^p(X) \subseteq L^r(X) + L^q(X)$.

Proof. Let $f \in L^p(X)$. For $a > 0$ write

$$f = \underbrace{f \cdot \chi_{\{|f(x)| > a\}}}_{f^a} + \underbrace{f \cdot \chi_{\{|f(x)| \leq a\}}}_{f_a}.$$

We now claim that $f^a \in L^r(X)$ and $f_a \in L^q(X)$. Indeed,

$$\int |f^a|^r d\mu = \int_{\{|f| > a\}} |f|^r d\mu = \int_{\{|f| > a\}} |f|^p \cdot \underbrace{|f|^{r-p}}_{\leq a^{r-p}} d\mu \leq \|f\|_p^p \cdot a^{r-p},$$

and

$$\int |f_a|^q d\mu = \int_{\{|f| \leq a\}} |f|^q d\mu = \int_{\{|f| \leq a\}} |f|^p \underbrace{|f|^{q-p}}_{\leq a^{q-p}} d\mu \leq \|f\|_p^p \cdot a^{q-p}.$$

□

Definition 10. Let U, V be vector spaces of functions from some set X to $\mathbb{C}$. A function $T : U \rightarrow V$ is said to be a *sublinear operator* if $|(T(f+g))(x)| \leq |(Tf)(x)| + |(Tg)(x)|$ and $|(T(cf))(x)| = |c| \cdot |(Tf)(x)|$ for all $f, g \in U$, $c \in \mathbb{C}$, $x \in X$.

Definition 11. Let $(X, \mathcal{A}, \mu)$, $(Y, \mathcal{B}, \nu)$ be measure spaces, let $0 < p < \infty$, and let $T : L^p(X) \rightarrow L^p(Y)$. We say that T is of *strong type* (p, p) if there exists a constant C_p with $\|Tf\|_{L^p(Y)} \leq C_p \|f\|_{L^p(X)}$ for all $f \in L^p(X)$. If instead there is a C_p with

$$\nu(\{y \mid |Tf(y)| > a\}) \leq \frac{C_p}{a^p} \|f\|_{L^p(X)}^p$$

for all $f \in L^p(X)$, $a > 0$, we say that T is of *weak type* (p, p) .

Remark 12. As the naming suggests, the strong type implies the weak type. Indeed, if T is of strong type (p, p) , then

$$\begin{aligned} \nu(\{y \mid |Tf(y)| > a\}) &= \frac{1}{a^p} \int_{\{|Tf|>a\}} a^p d\nu \\ &\leq \frac{1}{a^p} \int_{\{|Tf|>a\}} |Tf|^p d\nu \\ &\leq \frac{\|Tf\|_{L^p(Y)}^p}{a^p} \\ &\leq \frac{C_p}{a^p} \|f\|_{L^p(X)}^p. \end{aligned}$$

Theorem 13 (Marcinkiewicz interpolation theorem). *Let $(X, \mathcal{A}, \mu)$, $(Y, \mathcal{B}, \nu)$ be σ -finite measure spaces, let $0 < q < r \leq \infty$ and let $T : L^q(X) + L^r(X) \rightarrow \{\text{functions } Y \rightarrow \mathbb{C}\}$ be a sublinear operator. If T is of weak types (q, q) and (r, r) (unless $r = \infty$, in which case we want it to be of strong type (r, r)), then T is of strong type (p, p) for all $p \in (q, r)$.*

Proof. Let us first deal with the case $p < \infty$. First rewrite,

$$\begin{aligned} \|Tf\|_{L^p(Y)}^p &= \int |Tf(y)|^p d\nu(y) \\ &= \int \int_0^{|Tf(y)|} p\alpha^{p-1} d\alpha d\nu(y) \\ &= \int \int_0^\infty \chi_{\alpha \leq |Tf(y)|}(\alpha, y) p\alpha^{p-1} d\alpha d\nu(y) \\ &= \int_0^\infty p\alpha^{p-1} \int_{\{y \mid |Tf(y)| > \alpha\}} 1 d\nu(y) d\alpha \\ &= \int_0^\infty p\alpha^{p-1} \nu(\{y \mid |Tf(y)| > \alpha\}) d\alpha. \end{aligned}$$

Now, for $\alpha > 0$ decompose $f = \underbrace{f\chi_{|f|>\alpha}}_{f_\alpha} + \underbrace{f\chi_{|f|\leq\alpha}}_{f_\alpha}$. By sublinearity, we have

that for all $y \in Y$, $|Tf(y)| \leq |Tf_\alpha(y)| + |Tf_\alpha(y)|$. Therefore,

$$\{y \in Y \mid |Tf(y)| > \alpha\} \subseteq \{y \in Y \mid |Tf_\alpha(y)| > \alpha/2\} \cup \{y \in Y \mid |Tf_\alpha(y)| > \alpha/2\},$$

which means that

$$\nu(\{|Tf(y)| > \alpha\}) \leq \nu(\{|Tf^\alpha(y)| > \alpha/2\}) + \nu(\{|Tf_\alpha(y)| > \alpha/2\})$$

and

$$\begin{aligned} \|Tf\|_{L^p(Y)}^p &\leq \underbrace{\int_0^\infty p\alpha^{p-1}\nu(\{|y| \mid |Tf^\alpha(y)| > \alpha/2\}) d\alpha}_A \\ &\quad + \underbrace{\int_0^\infty p\alpha^{p-1}\nu(\{|y| \mid |Tf_\alpha(y)| > \alpha/2\}) d\alpha}_B. \end{aligned}$$

Because T was assumed to be of weak type (q, q) , we have that

$$\begin{aligned} A &\leq \int_0^\infty p\alpha^{p-1} \frac{C_q}{(\alpha/2)^q} \left(\int |f^\alpha|^q d\mu \right) d\alpha \\ &= 2^q p C_q \int_0^\infty \alpha^{p-q-1} \int_{\{|f|>\alpha\}} |f|^q d\mu d\alpha \\ &= 2^q p C_q \int |f(x)|^q \int_0^{|f(x)|} \alpha^{p-q-1} d\alpha d\mu(x) \\ &= \frac{2^q p C_q}{p-q} \int |f|^q |f|^{p-q} d\mu \\ &= \frac{2^q p C_q}{p-q} \|f\|_{L^p(X)}^p. \end{aligned}$$

Likewise,

$$\begin{aligned} B &\leq \int_0^\infty p\alpha^{p-1} \frac{C_r}{(\alpha/2)^r} \left(\int |f_\alpha|^r d\mu \right) d\alpha \\ &= 2^r p C_r \int_0^\infty \alpha^{p-r-1} \int_{\{|f|\leq\alpha\}} |f|^r d\mu d\alpha \\ &= 2^r p C_r \int |f(x)|^r \int_{|f(x)|}^\infty \alpha^{p-r-1} d\alpha d\mu(x) \\ &= \frac{2^r p C_r}{r-p} \int |f|^r |f|^{p-r} d\mu \\ &= \frac{2^r p C_r}{r-p} \|f\|_{L^p(X)}^p \end{aligned}$$

Combining these gives that

$$\|Tf\|_{L^p(Y)}^p \leq p \left(\frac{2^q C_q}{p-q} + \frac{2^r C_r}{r-p} \right) \|f\|_{L^p(X)}^p,$$

and taking p -th roots gives the strong type constant.

In the other case, i.e. $r = \infty$, the assumption is that $\|Tf\|_{L^\infty(Y)} \leq C_\infty \|f\|_{L^\infty(X)}$.

Decompose f as $f = \tilde{f}^\alpha + \tilde{f}_\alpha$ where $\tilde{f}^\alpha = f\chi_{|f| > \alpha/(2C_\infty)}$ and $\tilde{f}_\alpha = f\chi_{|f| \leq \alpha/(2C_\infty)}$.

Like in the first case, we have by sublinearity that,

$$|Tf(x)| \leq |T\tilde{f}^\alpha(x)| + |T\tilde{f}_\alpha(x)|$$

and

$$\nu(\{y \mid |Tf(y)| > \alpha\}) \leq \nu\left(\{y \mid |T\tilde{f}^\alpha(y)| > \alpha/2\}\right) + \nu\left(\{y \mid |T\tilde{f}_\alpha(y)| > \alpha/2\}\right).$$

The first term can be bounded like A in the first case, while the second term is zero by the definition of $\tilde{f}_\alpha$. Indeed,

$$\nu\left(\{y \mid |T\tilde{f}_\alpha(y)| > \alpha/2\}\right) \leq \nu\left(\{y \mid C_\infty |\tilde{f}_\alpha(y)| > \alpha/2\}\right) = \nu\left(\underbrace{\{y \mid |\tilde{f}_\alpha(y)| > \alpha/(2C_\infty)\}}_{=\emptyset}\right).$$

□

Corollary 14. *For any $1 < p < \infty$, M (i.e. the maximal function operator) is of strong type (p, p) .*

Proof. Theorem 7 and Remark 5 give the edge bounds for Theorem 13. □

Lecture 3

Theorem 15 (Lebesgue Differentiation theorem). *Let $f \in L^1_{loc}(\mathbb{R}^d)$, then*

$$\frac{1}{|B_r(x)|} \int_{B_r(x)} f(y) dy \rightarrow f(x)$$

as $r \rightarrow 0$ for almost every $x \in \mathbb{R}^d$.

Proof. We begin by letting by defining the operator R on $L^1_{loc}(\mathbb{R}^d)$

$$R(f)(x) = f_r(x) := \frac{1}{|B_r(x)|} \int_{B_r(x)} f(y) dy.$$

We then define the *oscillation* which is defined on $L^1(\mathbb{R}^d)$,

$$\text{osc}(f) = |\limsup_{r \rightarrow 0} f_r(x) - \liminf_{r \rightarrow 0} f_r(x)|.$$

It is easy to see that $\text{osc}(f)(x)$ is less or equal to $2M_c f(x)$ and therefore also $2Mf(x)$. Additionnaly it is easy to see that if f is continuous than the limit

$\lim_{r \rightarrow 0} f_r(x)$ exists and therefore the oscillation is zero. Now we note that if $f, h \in L^1(\mathbb{R}^d)$ and h continuous where $h_r(x) \rightarrow c(x)$ then

$$\begin{aligned} 2M(f+h)(x) &\geq \text{osc}(f+h) = |\limsup_{r \rightarrow 0} (f_r + h_r) - \liminf_{r \rightarrow 0} (f_r + h_r)| = \\ &= |\limsup_{r \rightarrow 0} \frac{1}{|B_r(x)|} \int_{B_r(x)} f(y) + h(y) dy - \liminf_{r \rightarrow 0} \frac{1}{|B_r(x)|} \int_{B_r(x)} f(y) + h(y) dy| = \\ &= |\limsup_{r \rightarrow 0} \frac{1}{|B_r(x)|} \int_{B_r(x)} f(y) dy + c(x) - \liminf_{r \rightarrow 0} \frac{1}{|B_r(x)|} \int_{B_r(x)} f(y) dy - c(x)| = \\ &= \text{osc}(f)(x). \end{aligned}$$

With this we aim at picking continuous h so that $2M(f+h)$ becomes small forcing $\text{osc}(f)(x)$ to zero. So pick arbitrary $\varepsilon > 0$ and continuous h such that $\|f+h\|_1 \leq \varepsilon$. Now as the maximal function is of weak type $(1, 1)$ we have that, for any $a > 0$,

$$\nu(\{y \mid |M(f+h)| > a\}) \leq \frac{C_p}{a^p} \|f+h\|_1 \leq \frac{C_p \varepsilon}{a^p}.$$

Since ε is arbitrary we get that for any $\delta > 0$ there exists an $\varepsilon > 0$ such that

$$\nu(\{\text{osc}(f) > \delta\}) \leq \varepsilon$$

. This gives that $\text{osc}(f)$ is zero almost everywhere. What we have proven is that the limit

$$\lim_{r \rightarrow 0} f_r(x)$$

exists almost everywhere and what remains is to show that it is equal to $f(x)$ almost everywhere. $\square$ **TODO**

Corollary 16. *For almost every $x \in \mathbb{R}^d$ we have that, for $f \in L^1_{loc}(\mathbb{R}^d)$, $|f(x)| \leq Mf(x)$. This gives that for each $p \in (1, \infty]$ there exists a $C_{p,d} > 0$ such that $\|f\|_p \leq \|Mf\|_p \leq C_{p,d} \|f\|_p$.*

Definition 17. A family $(k_t)_{t>0}$ of functions in $L^1(\mathbb{R}^d)$ is called an approximate identity if

- (i) $\int_{\mathbb{R}^d} k_t(x) dx = 1 \quad \forall t > 0$;
- (ii) $\exists C > 0 : \|k_t\|_1 < C \quad \forall t > 0$;
- (iii) $\forall \delta > 0 \quad \int_{|x|>\delta} |k_t(x)| dx \rightarrow 0$ as $t \rightarrow 0$.

Example 18. Let $k \in L^1(\mathbb{R}^d)$ with $\int_{\mathbb{R}^d} k(x) dx = 1$. Then we can define $k_t(x) := t^{-d} k(t^{-1}x)$ and this will be an approximate identity.

Theorem 19. *Let $(k_t)_{t>0}$ be an approximate identity, then*

(i) *if $f \in L^p(\mathbb{R}^d)$, $1 \leq p < \infty$, then*

$$\|k_t * f - f\|_{L^p} \rightarrow 0 \text{ as } t \rightarrow 0;$$

(ii) *if $f \in C_0(\mathbb{R}^d)$, then*

$$k_t * f \rightarrow f \text{ uniformly as } t \rightarrow 0.$$

Proof. For (i) we let $f \in L^p(\mathbb{R}^d)$, $1 \leq p < \infty$. Now we write

$$k_t * f - f = \int_{\mathbb{R}^d} k_t(x)(f(x-y) - f(x))dy$$

and note that if f was continuous we could split the integral into one of a sphere and one of an anti-sphere defined by some delta. $(f(x-y) - f(x))$ would be small in the sphere and $k_t(y)$ would be small in the anti-sphere and we would be done. However, of course f is not necessarily continuous and we cannot do this. Except we can, as we shall see.

Lemma 20. For $y \in \mathbb{R}^d$ and $1 \leq p < \infty$ let $T_y : L^p(\mathbb{R}^d) \rightarrow L^p(\mathbb{R}^d)$ send $f(x)$ to $f(x-y)$. Then for $f \in L^p(\mathbb{R}^d)$ we get

$$\|T_y f - f\|_p \rightarrow 0 \text{ as } y \rightarrow 0.$$

Proof. TODO □

Now we do

$$\begin{aligned} \|k_t * f - f\|_p &\leq \left(\int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |k_t(x)| |f(x-y) - f(x)| dy \right)^p dx \right)^{1/p} \leq \\ &\leq \int_{\mathbb{R}^d} |k_t(y)| \left(\int_{\mathbb{R}^d} |f(x-y) - f(x)|^p dx \right)^{1/p} dy \\ &= \int_{\mathbb{R}^d} |k_t(y)| \cdot \|T_y f - f\|_p dy \end{aligned}$$

The last inequality follows from Minkowski's inequality (apparently?). Now we can do the split up we wanted to do in the continuous case.

(ii) TODO (as in the Fourier analysis course apparently) □

We will now generalise the Lebesgue Differentiation Theorem, starting with an example.

Example 21. Let $k = V_d^{-1} \chi_{B_1(0)}$ where $V_d^{-1} = |B_1(0)|$ and to make an approximate identity define

$$k_t = t^{-d} k(t^{-1}x)$$

then we have that

$$\begin{aligned} M_c f(x) &= \sup_{t>0} \frac{1}{|B_t(x)|} \int_{B_t(x)} |f(y)| dy = \\ &= \sup_{t>0} \frac{1}{t^d V_d} \int_{B_t(0)} |f(x-y)| dy = \\ &= \sup_{t>0} t^{-d} V_d^{-1} \int_{\mathbb{R}^d} \chi_{B_1(0)}(t^{-1}y) \cdot |f(x-y)| dy = \\ &= \sup_{t>0} k_t * |f|. \end{aligned}$$

In other words the centered maximal function arises from convolution with a certain approximal distribution. This can be generalized.

Theorem 22. *Let g be a function from $[0, \infty)$ to $[0, \infty)$ which has at most finitely many discontinuities and is decreasing. Consider $k : \mathbb{R}^d \rightarrow \mathbb{R}$ given by $k(x) = g(|x|)$. If $k \in L^1(\mathbb{R}^d)$ then $\forall f \in L^1_{loc}(\mathbb{R}^d)$ one has*

$$\sup_{t>0} (|f| * k_t)(x) \leq \|k\|_1 M_c f$$

for almost every $x \in \mathbb{R}^d$.

No proof

Corollary 23. *Let g be as in the theorem above. Let further $k \in L^1(\mathbb{R}^d)$, $\int_{\mathbb{R}^d} k dx = 1$ and $|k(x)| \leq g(|x|)$. Then for $f \in L^p(\mathbb{R}^d)$, $p \in (1, \infty)$ we have*

$$k_t * f \rightarrow f(x) \text{ as } t \rightarrow 0$$

for almost every $x \in \mathbb{R}^d$.

How does this follow? TODO

We will now deal with the Schwartz class and it's dual.

Definition 24. The Schwartz Class denoted $\mathcal{S}(\mathbb{R}^d)$ encompasses all functions $f \in C^\infty(\mathbb{R}^d)$ satisfying

$$p_{a,b}(f) := \sup_{x \in \mathbb{R}^d} |x^a \partial^b f(x)| < \infty \quad \forall a, b \in \mathbb{N}_0^d.$$

Here a and b are multi-indices and $\partial^b = \partial_1^{b_1} \dots \partial_d^{b_d}$, $x^a = x_1^{a_1} \dots x_d^{a_d}$. We have in this space a metric induced by all the $p_{a,b}$ norms defined as

$$d(f, g) = \sum_{j=1}^{\infty} 2^{-j} \frac{p_j(f - g)}{1 + p_j(f - g)}$$

where (p_j) is an enumeration of all the $p_{a,b}$'s. See that we converge in this metric if and only if we converge in each $p_{a,b}$ norm.

This is a space with particularly nice functions. For example the Fourier transform maps the Schwartz space to itself.

Definition 25. A linear map $u : \mathcal{S}(\mathbb{R}^d) \rightarrow \mathbb{C}$ is called bounded if

$$u(\varphi_k) \rightarrow 0 \text{ as } k \rightarrow 0$$

whenever $\varphi_k \rightarrow 0$ in d .

Definition 26. The space of tempered distributions is defined as the dual of $\mathcal{S}(\mathbb{R}^d)$ which is the space of bounded linear maps $\mathcal{S}(\mathbb{R}^d) \rightarrow \mathbb{C}$. it is denoted

$$\mathcal{S}(\mathbb{R}^d)'.$$

Lecture 4

The space of tempered distributions is very big. For example note that for any $f \in L^1(\mathbb{R}^d)$ we may for $\varphi \in \mathcal{S}(\mathbb{R}^d)$ identify

$$\varphi \mapsto \langle f, \varphi \rangle = \int_{\mathbb{R}^d} f \cdot \bar{\varphi} dx$$

which defines a tempered distribution. Connecting tempered distributions to functions on $\mathbb{R}^d$ this way will be quite central so we will, for u a tempered distribution, write $\langle u, \varphi \rangle$ for $u(\varphi)$.

Example 27. Some examples of distributions:

- 1). The dirac delta function δ_0 , maps φ to $\varphi(0)$.
- 2). Any L^p function gives a distribution. 3). Define $p.v.(\frac{1}{x}) : \mathcal{S}(\mathbb{R}^d) \rightarrow \mathbb{C}$ by

$$\langle p.v.(\frac{1}{x}), \varphi \rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon \leq |x| \leq 1} \frac{\varphi(x)}{x} dx + \int_{|x| > 1} \frac{\varphi(x)}{x} dx.$$

This is a tempered distribution (p.v. stand for principal valued).

Definition 28. We will define what it means to differentiate a distribution. We do this through the connection to normal functions in the inner product. First recall that with partial integration we have for $\varphi, \psi \in \mathcal{S}(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} \partial^a \varphi(x) \psi(x) dx = \int_{\mathbb{R}^d} \varphi(x) \partial^a \psi(x) dx.$$